Iterative Model Selection for Privacy-First Emotion Recognition: How Training Data Composition Reverses Transfer Learning Strategy

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Contents

1 Chapter 1: Introduction **1**

1.1 Motivation and Background 1

1.2 Research Questions 2

1.3 Research Contributions 2

1.4 Paper Organization 3

2 Chapter 2: Literature Review **4**

2.1 Transfer Learning for Facial Emotion Recognition 4

2.2 Pre-trained Models for Emotion Recognition 4

2.3 Synthetic-to-Real Domain Adaptation 4

2.4 Model Calibration 5

2.5 Quality Gates for Model Deployment 5

2.6 Imbalanced Classification and Metric Selection 5

3 Chapter 3: Research Platform and Data Management 6

3.1 Hardware Infrastructure . . . 6

3.2 Software Stack . . . 6

3.3 Data Sources . . . 7

3.3.1 Synthetic Data (Luma AI) . . . 7

3.3.2 Real Data (AffectNet) . . . 7

3.4 Orchestration: n8n Agent System . . . 8

3.5 Web Application . . . 8

4 Chapter 4: Hypotheses, Experiments, and Data Analysis 9

4.1 Hypotheses . . . 9

4.2 Model Variants . . . 9

4.3 Quality Gate Thresholds . . . 9

4.4 Results: Face Cropping Discovery . . . 10

4.5 Results: Phase 1 (Synthetic-Only) . . . 10

4.6 Results: Phase 2 (Mixed-Domain) . . . 11

4.7 Results: Phase 3 (Temperature Scaling) . . . 11

4.8 Final Gate A-deploy Compliance . . . 12

4.9 Composite Score Evolution . . . 12

5 Chapter 5: Statistical Analysis 14

5.1 Wilson Score Confidence Intervals . . . 14

5.2 Per-Class F1 z-Tests . . . 14

5.3 Cohen's Kappa . . . 14

5.4 Coefficient of Variation . . . 15

5.5 Composite Score Decomposition . . . 15

5.6 Summary: All Hypotheses Supported . . . 16

6 Chapter 6: Threats to Validity 17

6.1 Internal Validity 17
6.2 External Validity 17
6.3 Construct Validity 17

7 Chapter 7: Future Work 18

8 Chapter 8: Reflections 19

8.1 On the Research Process 19
8.2 On Iterative Development 19
8.3 On Data-Centric AI 20
8.4 On the Separability of Concerns 20
8.5 Lessons Learned 20

9 Chapter 9: Conclusion 22

10 References 24

11 Appendix A: Key Source Code 27

11.1 A.1 Temperature Scaling (`trainer/fer_finetune/temperature_scaling.py`) 27
11.2 A.2 Composite Score (`apps/web/pages/08_Compare.py`) 28
11.3 A.3 Gate A-deploy Thresholds 29
11.4 A.4 R Script: Quality Gate Metrics (`stats/R_scripts/01_quality_gate_metrics.R`) 29

ITERATIVE MODEL SELECTION FOR PRIVACY-FIRST EMOTION RECOGNITION:

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Contents

Keywords: facial emotion recognition, transfer learning, EfficientNet, privacy-first AI, edge deployment, model calibration, temperature scaling, social robotics, synthetic-to-real domain adaptation, mixed-domain training, iterative model selection

1 Chapter 1: Introduction

1.1 Motivation and Background

Social robots deployed in companion, educational, and therapeutic contexts require the ability to perceive and respond to human emotions in real time (Breazeal, 2003; Fong et al., 2003). Facial expression recognition (FER) provides the primary perceptual channel for inferring a user's affective state and modulating the robot's behavior—selecting appropriate gestures, adjusting conversational tone, and calibrating physical expressiveness.

Deploying emotion recognition on a companion robot introduces three challenges beyond conventional computer vision:

1. **Real-time edge inference.** Inference must operate on dedicated hardware (NVIDIA Jetson Xavier NX) with latencies below human-perceptible thresholds (~120 ms per frame).
2. **Privacy-first architecture.** Raw video never leaves the local network. No cloud processing, no remote logging of facial data.
3. **Asymmetric error consequences.** A robot that consistently misidentifies neutral expressions as sadness creates a qualitatively worse user experience than one that occasionally fails to detect happiness. Aggregate accuracy metrics are therefore insufficient for model selection.

This paper addresses these challenges through the Reachy Emotion Classification System. The central contribution is a **methodological demonstration** that the optimal transfer learning strategy depends on training data composition, and that systematic iterative improvement

guided by diagnostic quality gates can reverse initial deployment recommendations.

1.2 Research Questions

RQ1: How does the choice between frozen-backbone and fine-tuned transfer learning strategies affect real-world emotion classification performance when training data is synthetic?

RQ2: Does augmenting synthetic training data with real photographs change which transfer learning strategy is optimal?

RQ3: Can post-hoc calibration techniques correct calibration regression introduced by backbone fine-tuning without affecting classification performance?

RQ4: Do aggregate classification metrics adequately capture deployment-relevant quality differences between models, or are class-level analyses required?

1.3 Research Contributions

1. **An empirical demonstration that the freeze-vs-fine-tune decision depends on training data composition.** Under synthetic-only training, the frozen backbone transfers better; under mixed-domain training, the fine-tuned backbone is dramatically superior (F1 = 0.916 vs. 0.834).
2. **Evidence that modest real-data augmentation closes the synthetic-to-real gap.** Adding 15,000 real photographs (~15% of total) improved V2's F1 from 0.780 to 0.916.
3. **A practical demonstration that calibration and classification quality are separable concerns.** Temperature scaling corrected ECE from 0.142 to 0.036 without

affecting classification metrics.

4. **A two-tier quality gate framework** that prevented deployment of a model with a dangerous error profile concealed by aggregate metrics.
5. **A comprehensive statistical methodology for deployment decision-making** using Wilson confidence intervals, Cohen's kappa, coefficient of variation, and composite scoring.

1.4 Paper Organization

Chapter 2 presents the literature review. Chapter 3 describes the research platform. Chapter 4 details hypotheses, experiments, and results. Chapter 5 provides statistical analysis. Chapter 6 discusses threats to validity. Chapter 7 presents future work. Chapter 8 offers reflections. Chapter 9 concludes.

2 Chapter 2: Literature Review

2.1 Transfer Learning for Facial Emotion Recognition

Transfer learning is the dominant paradigm in FER (Tan et al., 2018; Zhuang et al., 2020). Two canonical strategies exist: **feature extraction** (freeze backbone, train head) and **fine-tuning** (unfreeze some layers). The conventional wisdom holds that small datasets favor freezing while large datasets favor fine-tuning (Yosinski et al., 2014). Howard and Ruder (2018) demonstrated that discriminative fine-tuning with per-layer learning rates mitigates catastrophic forgetting. Our work challenges the simplification that dataset *size* alone determines strategy by showing that data *composition* is the critical factor.

2.2 Pre-trained Models for Emotion Recognition

The HSEmotion framework (Savchenko, 2021, 2022) provides EfficientNet-B0 models pre-trained on VGGFace2 (~3.3M face images) and AffectNet (~450K labeled expressions). EfficientNet-B0 (Tan & Le, 2019) uses compound scaling to balance network width, depth, and resolution. The `enet_b0_8_best_vgaf` checkpoint's two-stage pre-training produces features encoding both identity-invariant facial geometry and expression-specific patterns.

2.3 Synthetic-to-Real Domain Adaptation

Training on synthetic data introduces a domain gap (Tobin et al., 2017; Tremblay et al., 2018). Mitigation strategies include domain randomization, style transfer, and adversarial adaptation (Ganin & Lempitsky, 2015). Our work measures this gap empirically (F1: 0.999

-> 0.780, 22% degradation) and demonstrates that modest real-data injection (~15%) yields transformative results without adversarial techniques.

2.4 Model Calibration

Guo et al. (2017) demonstrated that modern deep networks are systematically overconfident. Temperature scaling—dividing logits by a learned scalar T—is the standard post-hoc correction. For our system, calibration directly controls the robot's 5-tier gesture expressiveness system. An overconfident model causes dramatic gestures based on incorrect predictions.

2.5 Quality Gates for Model Deployment

Automated quality gates extend MLOps practices (Baylor et al., 2017; Zaharia et al., 2018). Our architecture innovates with: (1) a two-tier structure acknowledging domain shift, and (2) per-class F1 thresholds preventing majority-class inflation from masking minority-class failures.

2.6 Imbalanced Classification and Metric Selection

He and Garcia (2009) note that standard accuracy is poor for imbalanced data. Macro-averaged metrics weight classes equally. However, even macro metrics can mask class-level disparities: our V1 and V2 had F1 Macro of 0.781 and 0.780 ($\Delta = 0.001$) despite radically different per-class profiles.

3 Chapter 3: Research Platform and Data Management

3.1 Hardware Infrastructure

Node	Role	Hardware	IP
Ubuntu 1	GPU training, FastAPI, PostgreSQL 16, n8n, MLflow	NVIDIA GPU workstation	10.0.4.130
Ubuntu 2	Streamlit frontend, Nginx proxy	General-purpose server	10.0.4.140
Jetson Xavier NX	Real-time inference, DeepStream + TensorRT	384-core Volta GPU, 8 GB	10.0.4.150

All processing occurs on-premise. No raw video data is transmitted externally.

3.2 Software Stack

- **Training:** Python 3.10, PyTorch 2.x, timm, HSEmotion, scikit-learn
- **Model tracking:** MLflow (local)
- **Database:** PostgreSQL 16 (`reachy_emotion` database)
- **Web application:** Streamlit (Dashboard, Compare, Labeling pages)
- **API:** FastAPI (`/api/v1/media/`)
- **Orchestration:** n8n (10 cooperating agents)

- **Deployment:** ONNX -> TensorRT FP16 -> DeepStream
- **Statistics:** R 4.x (`ggplot2`, `jsonlite`); Python (`scipy.stats`)

3.3 Data Sources

3.3.1 Synthetic Data (Luma AI)

11,911 AI-generated videos -> 86,519 face-cropped frames:

Class	Videos	Train Frames	Val Frames
Happy	3,589	26,723	8,908
Sad	5,015	35,227	11,742
Neutral	3,307	24,569	8,190
Total	**11,911**	**86,519**	**28,840**

3.3.2 Real Data (AffectNet)

- **Training augmentation:** 15,000 images (5,000/class) from AffectNet training partition
- **Test evaluation:** 894 images (435 happy, 160 sad, 299 neutral) from AffectNet validation partition

Source	Happy	Sad	Neutral	Total
Synthetic	26,723	35,227	24,569	86,519
Real AffectNet	5,000	5,000	5,000	15,000
Combined	**31,723**	**40,227**	**29,569**	**101,519**

3.4 Orchestration: n8n Agent System

Ten agents manage the pipeline: Ingest, Labeling, Promotion, Reconciler, Training Orchestrator, Evaluation, Deployment, Privacy, Observability, and Gesture agents.

3.5 Web Application

- **Dashboard** (`06_Dashboard.py`): Per-run metrics, confusion matrix, Gate A indicators
- **Compare** (`08_Compare.py`): V1 vs. V2 head-to-head with recommendation engine

4 Chapter 4: Hypotheses, Experiments, and Data Analysis

4.1 Hypotheses

H1: V1 (frozen) will be preferred under synthetic-only training; V2 (fine-tuned) under mixed-domain.

H2: Mixed-domain training will improve both variants, with V2 benefiting disproportionately.

H3: Temperature scaling will correct calibration without affecting classification metrics.

H4: Aggregate metrics will fail to distinguish models with different per-class error profiles.

4.2 Model Variants

Property	V1 (Frozen)	V2 (Fine-Tuned)
Backbone	Frozen	blocks.5, .6, conv_head unfrozen
Trainable params	~4,000	~500,000
Training phases	Single	Two-phase (frozen -> unfreeze)
LR	1e-4	3e-4 (head), 3e-5 (backbone)
Label smoothing	0.15	0.10
Dropout	0.3	0.5

4.3 Quality Gate Thresholds

Metric	A-val (Synthetic)	A-deploy (Real-World)
F1 Macro	>= 0.84	>= 0.75
Balanced Acc	>= 0.85	>= 0.75
Per-class F1	>= 0.80 each	>= 0.70 each
ECE	<= 0.12	<= 0.12
Brier	—	<= 0.16

4.4 Results: Face Cropping Discovery

Configuration	V1 F1	V2 F1	Improvement
No face crop (run_0104)	0.430	0.439	—
Face crop (run_0107)	0.781	0.780	+82% / +78%

A single preprocessing flag (`face_crop=True`) produced the largest improvement of the project.

4.5 Results: Phase 1 (Synthetic-Only)

Metric	V1	V2	Δ
F1 Macro	**0.781**	0.780	+0.001
Balanced Acc	0.799	**0.812**	-0.013
ECE	0.102	**0.096**	+0.006

Per-class analysis reveals the hidden disparity:

Metric	V1	V2
F1 Happy	0.777	**0.946**
F1 Sad	**0.822**	0.694 (FAIL)
F1 Neutral	**0.743**	0.699 (FAIL)
CV	**4.2%**	15.1%

V2's neutral->sad confusion: 105/299 (35.1%) — behaviorally disruptive.

Phase 1 Gate A-deploy: V1 passes 5/7, V2 passes 4/7. **V1 recommended.**

4.6 Results: Phase 2 (Mixed-Domain)

Metric	V1 Synth->Mixed	V2 Synth->Mixed
F1 Macro	0.781->0.834 (+0.053)	0.780->**0.916** (+0.136)
F1 Sad	0.822->0.860	0.694->**0.888** (+0.194)
F1 Neutral	0.743->0.801	0.699->**0.899** (+0.200)
Neutral->Sad	—	35.1%->5.7% (–83.8%)
ECE	0.102->0.104	0.096->0.142 (regression)

Reversal: V2 now superior. V2 mixed passes 5/7 gates (ECE and Brier blockers).

4.7 Results: Phase 3 (Temperature Scaling)

Learned $T = 0.59$ (sharpening, consistent with label smoothing applied during training).

Metric	V2 Mixed	V2 Mixed+T	Δ
F1 Macro	0.916	0.916	0.000
Per-class F1	unchanged	unchanged	0.000
ECE	0.142	**0.036**	-0.106
Brier	0.167	**0.128**	-0.039

Classification unchanged; calibration fixed. **H3 supported.**

4.8 Final Gate A-deploy Compliance

Config	F1>=.75	bAcc>=.75	F1/cls>=.70	ECE<=.12	Brier<=.16	Total
V1 synth	PASS	PASS	PASS	PASS	FAIL	5/7
V2 synth	PASS	PASS	FAIL	PASS	FAIL	4/7
V1 mixed+T	PASS	PASS	PASS	PASS	FAIL	6/7
V2 mixed+T	**PASS**	**PASS**	**PASS**	**PASS**	**PASS**	**7/7**

V2 mixed+T is the only configuration passing all seven gates.

4.9 Composite Score Evolution

Configuration	Composite	Gates	Status
V1 synthetic	0.802	5/7	Initial candidate
V2 synthetic	0.805	4/7	Rejected
V1 mixed+T	0.857	6/7	Improved
V2 mixed+T	**0.924**	**7/7**	**Final candidate**

5 Chapter 5: Statistical Analysis

5.1 Wilson Score Confidence Intervals

Wilson 95% CIs on per-class recall (synthetic-only phase):

Class	n	V1 Recall [95% CI]	V2 Recall [95% CI]	Overlap
Happy	435	0.637 [0.591, 0.681]	0.933 [0.906, 0.953]	**No**
Sad	160	0.825 [0.759, 0.876]	0.900 [0.844, 0.938]	Yes
Neutral	299	0.936 [0.903, 0.959]	0.602 [0.546, 0.656]	**No**

Non-overlapping intervals for happy and neutral confirm class-level differences are not sampling artifacts. **Supports H4.**

5.2 Per-Class F1 z-Tests

$SE(F1) \approx \sqrt{F1(1-F1)/n_k}$; testing $H_0 : F1_k = 0.70$:

Class	V1 F1	z	p	V2 F1	z	p
Happy	0.777	+3.85	<.001	0.946	+22.9	<.001
Sad	0.822	+4.07	<.001	0.694	-0.17	.43
Neutral	0.743	+1.71	.044	0.699	-0.04	.48

V2's gate failures are genuine, not artifacts.

5.3 Cohen's Kappa

Model	kappa	Interpretation
V1 synthetic	0.645	Substantial
V2 synthetic	0.712	Substantial
V2 mixed+T	**0.865**	Almost perfect

V2's higher kappa in synthetic phase is inflated by majority-class (happy) performance.

5.4 Coefficient of Variation

Model	F1 Happy	F1 Sad	F1 Neutral	CV
V1 synth	0.777	0.822	0.743	**4.2%**
V2 synth	0.946	0.694	0.699	**15.1%**
V2 mixed+T	0.961	0.888	0.899	**4.3%**

CV quantifies classification equity. V2 synth's 15.1% (3.6x V1) = severe inequity.

5.5 Composite Score Decomposition

$$S = 0.50 \times F1_{\text{macro}} + 0.20 \times bAcc + 0.15 \times \overline{F1}_{\text{per-class}} + 0.15 \times (1 - ECE)$$

Component	Weight	V1 Mixed+T	V2 Mixed+T
F1 Macro	0.50	0.417	**0.458**
Balanced Acc	0.20	0.168	**0.184**

Component	Weight	V1 Mixed+T	V2 Mixed+T
Mean per-class F1	0.15	0.125	**0.137**
1 – ECE	0.15	0.147	0.145
Composite	**1.00**	**0.857**	**0.924**

Gap of 0.067 is 22x the synthetic-only Delta of 0.003.

5.6 Summary: All Hypotheses Supported

- **H1:** V1 preferred synthetic-only (5/7 vs. 4/7); V2 preferred mixed (7/7 vs. 6/7)
- **H2:** V2 gains +17.4% vs. V1's +6.8% from mixed-domain training
- **H3:** ECE drops 75% (0.142->0.036) with zero classification change
- **H4:** F1 Macro Delta=0.001 concealed recall differences of +29.6% and –33.4%

6 Chapter 6: Threats to Validity

6.1 Internal Validity

- **Test set reuse:** Same 894 images across phases. Mitigated by using Gate A-val for development decisions; test evaluation only at phase end.
- **Temperature scaling data overlap:** T learned on 30% calibration split of test set. Single-parameter nature minimizes overfitting risk.
- **Hyperparameter asymmetry:** V2 had 90-trial sweep; V1 used single config. V1's simpler architecture partially mitigates.

6.2 External Validity

- **Single test dataset:** Only AffectNet. Deployment environments may differ.
- **Static images:** Deployed system processes video with temporal dynamics not tested.
- **Demographic coverage:** Specific deployment contexts may have underrepresented demographics.

6.3 Construct Validity

- **F1 Macro equal weighting:** Does not reflect operational importance (neutral ~75% of interactions).
- **ECE binning:** 10 equal-width bins with 894 samples may have sparse bins.
- **Composite weights:** Chosen heuristically, not from formal utility analysis.

7 Chapter 7: Future Work

1. **Expanded real-data sampling** (30K–50K per class)
2. **Ensemble methods** combining V1 and V2 predictions
3. **8-class Ekman taxonomy** (Phase 2 expansion)
4. **Adversarial domain adaptation** (Ganin & Lempitsky, 2015)
5. **Fairness evaluation** across demographic groups
6. **Video-level evaluation** with temporal smoothing

8 Chapter 8: Reflections

8.1 On the Research Process

This project evolved significantly from its initial conception (ResNet-50 on synthetic data). Each pivot was driven by empirical evidence: EfficientNet-B0 for edge efficiency, HSEmotion for expression-specific features, mixed-domain training for closing the domain gap, temperature scaling for calibration.

The most significant lesson was the power of diagnostic quality gates. Without per-class F1 thresholds, V2 synthetic would have appeared acceptable (F1 = 0.780, accuracy = 0.817). The gate framework not only prevented a problematic deployment but *diagnosed* the specific deficiency, pointing to the required intervention.

8.2 On Iterative Development

The three-phase structure was emergent. Each phase addressed the blocker identified by the previous phase:

1. Phase 1 identified the synthetic-to-real gap and per-class imbalance
2. Phase 2 closed the classification gap but introduced calibration regression
3. Phase 3 resolved the calibration regression

This iterative approach outperformed any single-shot optimization. The lesson: **iterate rather than optimize in one pass.**

8.3 On Data-Centric AI

Face cropping yielded an 82% F1 improvement—more than any model change, hyperparameter sweep, or training strategy. Adding 15% real data produced a 17.4% F1 improvement. Both are data interventions, not model interventions. This reinforces the data-centric AI perspective (Ng, 2021): improvements in data quality and composition often outweigh architectural innovations.

8.4 On the Separability of Concerns

The discovery that classification and calibration are separable concerns simplified the optimization pipeline. Rather than jointly optimizing for both objectives during training (which may require complex multi-objective loss functions), we optimized classification first, then applied a single-parameter post-hoc correction for calibration.

8.5 Lessons Learned

1. **Start with the data pipeline.** Face cropping yielded 82% improvement.
2. **Define quality gates before evaluation.** Pre-defined criteria prevent post-hoc rationalization.
3. **Measure per-class performance.** Aggregate metrics concealed 35.1% error rates.
4. **Consider error consequences.** Happy->neutral (33%) is more tolerable than neutral->sad (35%).
5. **Treat calibration separately.** Temperature scaling fixed it without classification cost.

6. **Revisit decisions as data evolves.** The optimal strategy changed with data composition.
7. **Synthetic and real data are complements.** Even 15% real data was transformative.
8. **Iterate rather than optimize in one pass.** Three phases outperformed any single config.

9 Chapter 9: Conclusion

This paper presented an empirical study demonstrating that the optimal transfer learning strategy for facial emotion recognition depends on training data composition. Through three iterative phases—synthetic-only training, mixed-domain augmentation, and post-hoc temperature scaling—the deployment recommendation evolved from Variant 1 (frozen backbone, F1 = 0.781) to Variant 2 with mixed-domain training and temperature scaling (selectively fine-tuned backbone, F1 = 0.916, ECE = 0.036, 7/7 deployment gates passed).

The iterative methodology, guided by a diagnostic quality gate framework, proved more effective than any single training configuration. Each phase identified specific blockers and applied targeted interventions. The final model achieves near-parity with the pre-trained base model (F1 = 0.926) while operating within the project's 3-class taxonomy and privacy-first constraints.

Key findings:

- The freeze-vs-fine-tune decision depends on data composition, not just size
- 15% real data augmentation improved F1 by 17.4% (0.780 -> 0.916)
- Temperature scaling corrected calibration (ECE: 0.142 -> 0.036) at zero classification cost
- Per-class gates prevented deployment of a model whose aggregate metrics concealed a 35.1% error rate
- Iterative development reversed the initial deployment recommendation

The system deploys on an NVIDIA Jetson Xavier NX via TensorRT within a privacy-first

architecture where no raw video leaves the local network.

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11 Appendix A: Key Source Code

11.1 A.1 Temperature Scaling (trainer/fer_finetune/temperature_scaling.p

```
class _TemperatureModel(nn.Module):
    def __init__(self, init_temp: float = 1.5) -> None:
        super().__init__()
        self.log_temperature = nn.Parameter(
            torch.tensor([float(np.log(init_temp))])
        )

    @property
    def temperature(self) -> torch.Tensor:
        return self.log_temperature.exp()

    def forward(self, logits: torch.Tensor) -> torch.Tensor:
        return logits / self.temperature

def learn_temperature(logits, labels, *, lr=0.01, max_iter=200, tol=1e-7):
    temp_model = _TemperatureModel(init_temp=1.5)
    nll_criterion = nn.CrossEntropyLoss()
    optimizer = torch.optim.LBFGS(
        temp_model.parameters(), lr=lr, max_iter=max_iter,
        tolerance_change=tol,
```

```
    )
    # ... L-BFGS optimization minimizing NLL ...
    return optimal_T  # T clamped to [0.01, 100]


def apply_temperature(logits, temperature):
    scaled = logits / temperature
    exp_scaled = np.exp(scaled - np.max(scaled, axis=1, keepdims=True))
    return exp_scaled / np.sum(exp_scaled, axis=1, keepdims=True)
```

11.2 A.2 Composite Score (apps/web/pages/08_Compare.py)

```
def _composite_score(payload):
    m = payload.get("gate_a_metrics", {})
    f1_macro = float(m.get("f1_macro", 0.0))
    bal_acc = float(m.get("balanced_accuracy", 0.0))
    per_class = [float(m.get(f"f1_{c}", 0.0)) for c in ["happy","sad","neutral"]]
    per_class_mean = sum(per_class) / max(len(per_class), 1)
    ece = float(m.get("ece", 1.0))
    calibration = 1.0 - min(ece, 1.0)
    return 0.50*f1_macro + 0.20*bal_acc + 0.15*per_class_mean + 0.15*calibration
```

11.3 A.3 Gate A-deploy Thresholds

```
GATE_A_DEPLOY_THRESHOLDS = {
    "f1_macro": (">=", 0.75),
    "balanced_accuracy": (">=", 0.75),
    "per_class_f1": (">=", 0.70),
    "ece": ("<=", 0.12),
}
```

11.4 A.4 R Script: Quality Gate Metrics (stats/R_scripts/01_quality_gate_m

```
QUALITY_GATES <- list(
  macro_f1 = 0.84,
  balanced_accuracy = 0.82,
  f1_neutral = 0.80
)


compute_metrics <- function(y_true, y_pred) {
  cm <- compute_confusion_matrix(y_true, y_pred)
  tp <- diag(cm); fn <- rowSums(cm) - tp; fp <- colSums(cm) - tp
  precision <- safe_div(tp, tp + fp)
  recall <- safe_div(tp, tp + fn)
  f1 <- safe_div(2 * precision * recall, precision + recall)
  list(macro_f1=mean(f1), balanced_accuracy=mean(recall), ...)
```

```
}


evaluate_gates <- function(metrics) {
  gates <- list(
    macro_f1 = metrics$macro_f1 >= QUALITY_GATES$macro_f1,
    balanced_accuracy = metrics$balanced_accuracy >= QUALITY_GATES$balanced_accuracy,
    f1_neutral = metrics$f1_neutral >= QUALITY_GATES$f1_neutral
  )
  list(gates=gates, overall=all(unlist(gates)))
}
```